

Euclidean algorithm and gcd



Theorem 1: (the division algorithm)

Let a, b be integers and $a > 0$. There

exists integers q, r such that $b = qa + r$

where $0 \leq r < a$. If $b = q'a + r'$

where $0 \leq r' < a$, then $q = q'$ and $r = r'$.

Lemma 2: (Archimedean property)

Let $x \in \mathbb{R}$. Then exists some $N \in \mathbb{N}$

such that $x \leq N$.

Corollary 3: Let $x, y \in \mathbb{R}$. $x > 0$.

Then $\exists$ some $N \in \mathbb{N}$ s.t. $y \leq xN$

proof of theorem 1.

Consider the set $R = \{ b - na : n \in \mathbb{Z} \} \cap \mathbb{N}$

We claim $R \neq \emptyset$. Indeed, by the

Archimedean property, $\exists$ some $N \in \mathbb{N}$

s.t. $-b \leq Na$ which is equivalent

to $b + Na \geq 0$. Let $n = -N$, we

see $b - na \geq 0$ and $R \neq \emptyset$.

By the Well-ordering Principle,

there is a least element in R

say $r = b - qa \geq 0$.

If $r \geq a$, then $r > r - a = b - (q+1)a \geq 0$

and $r - a$ is an element in R

that is smaller than the least element r .

This is a contradiction. So $r < a$.

We conclude the existence.

Uniqueness can be showed similarly.

Corollary 4: Every integer a can be written either as $a = 2k$ or $a = 2k + 1$.

i.e., an integer is either even or odd.

The division algorithm produces the remainder after dividing b by a . In fact, it can produce something more.

Def 5. Let $a, b > 0$ be integers.

The greatest common divisor is the largest integer n that divides both a, b , which we denote by (a, b) .

Lemma 6: For integers $0 < a < b$.

$$(b, a) = (a, b-a).$$

proof. Let $d = (b, a)$, $d' = (a, b-a)$

Since $d \mid a$, $d \mid b$, then $d \mid b-a$.

and hence $d \mid d'$ and $d \leq d'$. Similarly

since $d' \mid a$, $d' \mid b-a$,

we get $d' \mid b = a + b - a$

so $d' \mid d$ and $d' \leq d$. So $d = d'$.

Cor 7. Let $0 < a < b$ and

by division algorithm. $b = qa + r$ for $0 \leq r < a$

Then $(b, a) = (a, r)$

Cor 8: Let $0 < a < b$. By performing the division algorithm repeatedly,

we will get finitely many equations

$$b = q_0 a + r_0$$

$$a = q_1 r_0 + r_1$$

$$r_0 = q_2 r_1 + r_2$$

$\vdots$

$$r_{n-1} = q_{n+1} r_n + r_{n+1}$$

such that $0 = r_{n+1} < r_n < \dots < r_0$

and $r_n = (a, b)$

That is, repeated performing the division algorithm produces greatest common divisor. This process is also called the Euclidean algorithm.

Recall that we characterized the remainder as the least element of a set

We do the same for the gcd.

Theorem 9. (Bezout's theorem)

Suppose $0 < a < b$ and $(a, b) = d$.

Then d is the smallest positive number

in the set $\{ax+by : x, y \in \mathbb{Z}\}$.

proof: let $S = \{ax+by : x, y \in \mathbb{Z}\} \cap \mathbb{N}_+$

Note $a, b \in S$ and so $S \neq \emptyset$.

By the well-ordering principle

S contains a least (positive) element.

say $d' = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$.

Then $d \mid d'$ and $d \leq d'$.

But the Euclidean algorithm implies

that we may express d in the

form of $ax + by$ for some $x, y \in \mathbb{Z}$.

So $d \in S$ and $d \leq d'$.

Then $d = d'$.

Cor 10: Let $0 < a < b$ be integers.

and d' be a common divisor of a, b .

Then $d' \mid d$.

Def 11. Let $a, b > 0$ be integers.

We say a, b are relatively prime if

$$(a, b) = 1.$$

Lemma 12 : (Euclid's Lemma)

Suppose $a, b > 0$ are relatively prime.

If $a \mid bc$, then $a \mid c$.

proof. Since $(a, b) = 1$,

by Bezout's theorem, $\exists x, y \in \mathbb{Z}$

s.t. $ax + by = 1$. This implies

$$axc + byc = c.$$

Now $a \mid axc$ and $a \mid byc$

So $a \mid axc + byc = c$

Cor 13: (Rational root test) Let $n > 0$

and $a_0, \dots, a_n \in \mathbb{Z}$ and $a_n \neq 0$

Let r/s be relatively prime.

Suppose r/s is a solution

to the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = 0$

Then. $s \mid a_n$ and $r \mid a_0$.

proof. Plugging in r/s , we get

$$a_n \frac{r^n}{s^n} + a_{n-1} \frac{r^{n-1}}{s^{n-1}} + \dots + a_0 = 0$$

$$a_n r^n + s a_{n-1} r^{n-1} + \dots + s^n a_0 = 0$$

$$\text{So } a_n r^n = -s (a_{n-1} r^{n-1} + \dots + s^{n-1} a_0)$$

$$s^n a_0 = -r (a_n r^{n-1} + \dots + s^{n-1} a_1)$$